NP- : , ,

2026-06-06

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1 :

(polynomial-time), n $O(n^k)$. **tractable( )**, k $O(n^k)$ **intractable( )** .

“ ” .

- **(undecidable):** (Halting Problem) .
- **(intractable):** k $O(n^k)$, (superpolynomial) .

NP- . : $\Theta(n^{100})$, tractable . .

2

, NP- .

(P)	(NP-complete)
: $O(\lvert V\rvert\,\lvert E\rvert)$	
: $O(E)$.	
2-CNF :	3-CNF

(, 2 3) .

2.1 vs

LONG-PATH ( ): $G = (V, E)$ k , k ?

2.1.1 NP- —

$|V|$, $|V| - 1$. “ $|V| - 1$ ” “ $|V| - 1$ ”, .

. HAM-PATH $\leq_P$ LONG-PATH.

. $G = (V, E)$ G $k = |V| - 1$ ( ).

(→) G $|V|$ $|V| - 1$ LONG-PATH “ ”.

(←) G $\geq |V| - 1$, $|V| - 1$ $|V| - 1$ → → . ■

HAM-PATH NP- LONG-PATH NP-hard , ( ) NP . **NP-** .

2.1.2 — “ ”

- **:** . “ v ” , Dijkstra · Bellman-Ford DP/ .
- **:** “v ” . v ( ). DP “ ”($2^{|V|}$) .

: , DAG DP . “ + ” .

2.2 vs

	Euler tour	Hamiltonian cycle
		( )
	$O(E)$	NP-

: **Euler tour , Hamiltonian cycle .**

2.2.1 Euler tour

Euler tour :

$$\forall v: \text{ in-degree}(v) = \text{out-degree}(v).$$

$O(V + E)$.

: v , v t in-edge t out-edge t . in-degree = out-degree.

(,): .

( ). v (out-edge). v in-edge out-edge 1 . in-degree(v) = out-degree(v) out-edge . . ■

, (Hierholzer, $O(E)$). " " .

2.2.2 Hamiltonian cycle

" " . — . 3-CNF-SAT NP- .

: Euler tour Hamiltonian cycle . .

2.3 2-CNF vs 3-CNF

(satisfiable) 0/1 1 . k-CNF k OR (clause) AND . 2-CNF , 3-CNF NP- .

3 : P, NP, NPC

3.1 P

. $O(n^k)$.

3.2 NP

(verifiable) . **(certificate)**, . .

3.2.1

$\langle v_1, \ldots, v_{|V|} \rangle$. : (1) V , (2) $(v_{|V|}, v_1)$.

	$O(\|V\|)$	$O(\|V\| + \|E\|)$
	$O(\|V\|)$	$O(\|V\|)$
	$O(\|V\|^2)$	$O(\|V\| + \|E\|)$

- **:** boolean $O(|V|)$, $O(|V|)$.
- **:** $O(1)$ $O(|V|)$, $O(|V| + |E|)$.

$O(|V|)$ $O(|V|^2)$ **-** . , NP . (, NP- .)

3.3 P NP

P . $P \subseteq NP$. ,

$$P \stackrel{?}{=} NP.$$

$P \neq NP$.

3.4 NPC

NP NP . :

NP- , NP ($P = NP$).

3.4.1 " "

: "NPC ?" **NP- ' '** .

- NP- "NP , NP " , .
- "intractable" **($P \neq NP$)** .
- NP-hard (Section 7.1).

: NP , NPC . " " . " " . " (NPC)" .

4 P vs NP

2026 P NP . **NP- .** 2000 100 . $P = NP$ $P \neq NP$.

· · NP- , · · . .

5 vs

( ) , NP- " / " . k .

- SHORTEST-PATH( ): u v .
- PATH( ): u v $\leq k$?

. SHORTEST-PATH k PATH . . .

6 (Polynomial-time Reduction)

6.1

A α B β .

1. .
2. α " " $\iff$ β " " (**iff**).

$A \leq_P B$, **"A B "** .

$A \to B$ ( ), "B A ", $A \leq B$.

6.2

- **1 ($\leq$):** $A \leq_P B$ $B \in P$ $A \in P$.
- **2 ($\leq$):** $A \leq_P B$ A NP-hard B NP-hard.

6.3 (direction)

X .

		X
X $(X \in P)$	C $X \leq_P C$	( )
X (X NP-hard)	A $A \leq_P X$	( )

NP- . :

(reduce FROM known-hard TO yours).

6.3.1 ( )

VERTEX-COVER NP-hard CLIQUE $\leq_P$ VERTEX-COVER .

. VC . CLIQUE $\leq_P$ VC CLIQUE VC CLIQUE . CLIQUE NP- ( ) . **.** VC . ■

(VC $\leq_P$ CLIQUE) "CLIQUE VC " , VC . .

6.4

$\leq_P$:

$$X \leq_P Y \quad Y \leq_P Z \implies X \leq_P Z,$$

( ). .

NP- . $A' \in NP$ $A' \leq_P A$, **NP-** A'' $A'' \leq_P A$:

$$A' \leq_P A'' \leq_P A \quad (\forall A' \in NP).$$

6.4.1

$A' \leq_P A'' \leq_P B \leq_P A$ (A'' NPC, A' NP):

- $A' \leq_P A \to A$ **NP-hard** (B).
- $A' \leq_P A'' \leq_P B \to A' \leq_P B \to B$ **NP-hard**.

. anchor A'' (A') " "( ) , anchor (B, A) NP-hard . $\leq_P$.

, B NP- $B \in NP$. B " NPC" — .

6.5

($P \neq NP$). **·** , .

1. **:** "B A ." $P = NP$.
2. **intractability + :** B NP- . . . (" " " " .)
3. **( ):** (: $\to$, $\to$).
4. **NP :** NP- " " — , .

7 NP-hard / NP-complete

- A **NP-hard**: $A' \in NP$ $A' \leq_P A$.
- A **NP-complete**: $A \in NP$ A NP-hard.

" " . "hard " .

7.1 Theorem (34.4)

NP- $P = NP$.

. $L \in P$ $L \in NPC$. $L' \in NP$ NP- $L' \leq_P L$. $L \in P$ 1 $L' \in P$. $NP \subseteq P$, $P \subseteq NP$ $P = NP$. ■

: NP NP- . "NP- = intractability " .

8

NP- .

1. $X \in NP$**:** .
2. X **NP-hard:** NP- Y $Y \leq_P X$.
   - (a) f + .
   - (b) **(→) :** yes yes (yes-).
   - (c) **(←) :** yes yes (yes).

8.1 "no no"

$\alpha \in A \iff f(\alpha) \in B$.

$$P \leftrightarrow Q \quad \equiv \quad \neg P \leftrightarrow \neg Q$$

"yes ↔ yes" "no ↔ no" ( ).

, no- :

	yes-	= no-
(→)	C SAT ⇒ ϕ SAT	ϕ UNSAT ⇒ C UNSAT
(←)	ϕ SAT ⇒ C SAT	C UNSAT ⇒ ϕ UNSAT

C UNSAT $\iff$ ϕ UNSAT, no ↔ no . **(←)** "C UNSAT ⇒ ϕ UNSAT" " " . no iff .

9 NP-

NP- .

$$\text{CIRCUIT-SAT} \rightarrow \text{SAT} \rightarrow \text{3-CNF-SAT} \rightarrow \text{CLIQUE} \rightarrow \text{VERTEX-COVER}$$

9.1 CIRCUIT-SAT — NP-

: AND/OR/NOT (1)?

- **NP:** . , 1 → .
- **NP-hard (Lemma 34.6):** (NP). .

9.2 SAT — CIRCUIT-SAT $\leq_P$ SAT

: ϕ ?

- **NP:** .
- **:** C x_i x_i , ϕ .

($\rightarrow$) C 1. 1 , ϕ 1.

($\leftarrow$) ϕ 1 1. , C 1 . ■

9.2.1 :

$\rightarrow$ **("p q"):** $1 \rightarrow 0$, ( ).

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

: $p \rightarrow q \equiv \neg p \vee q$. " " — p .

$\leftrightarrow$ **(" "):** $p \leftrightarrow q$ 1.

. $\phi = ((x_1 \rightarrow x_2) \vee \neg(\neg x_1 \leftrightarrow x_3) \vee x_4) \wedge \neg x_2$, $x_1{=}0, x_2{=}0, x_3{=}1, x_4{=}1$.

- $x_1 \rightarrow x_2 = 0 \rightarrow 0 = 1$ ( ).
- $\neg(\neg x_1 \leftrightarrow x_3) = \neg(1 \leftrightarrow 1) = \neg 1 = 0$.
- $x_4 = 1$.
- OR: $1 \vee 0 \vee 1 = 1$.
- $\neg x_2 = 1$.
- $\phi = 1 \wedge 1 = 1 \rightarrow$.

$0 \rightarrow 0 = 1$.

9.3 3-CNF-SAT — SAT $\leq_P$ 3-CNF-SAT (Theorem 34.10)

3-CNF: 3 OR AND. : $(x_1 \vee \neg x_1 \vee \neg x_2) \wedge (x_3 \vee x_2 \vee x_4) \wedge (\neg x_1 \vee \neg x_3 \vee \neg x_4)$.

- **NP:** SAT NP .
- **:** ϕ 3-CNF ϕ' ($\rightarrow$ 3 $\rightarrow$).

Theorem 34.10. 3-CNF NP- .

9.4 CLIQUE — 3-CNF-SAT $\leq_P$ CLIQUE

: $G = (V, E)$ ( ). **:** k ?

- **NP:** V' $\rightarrow$.

. $\phi = C_1 \wedge ... \wedge C_k$, $C_r = (l_1^r \vee l_2^r \vee l_3^r)$.

- **:** C_r v_1^r, v_2^r, v_3^r ($3k$).
- **:** v_i^r v_j^s $\iff$ (1) ($r \neq s$) (2) (l_i^r l_j^s).

k. $O((3k)^2)$.

($\rightarrow$) ϕ 1 . $V'(k)$. V' $v_i^r, v_j^s (r \neq s)$ 1 (1), $\rightarrow$ $\rightarrow V'$.

($\leftarrow$) G k V' , V' . $v_i^r \in V'$ 1 , $\rightarrow \phi$.
$\blacksquare$

: (,) , NP-hard .

9.5 VERTEX-COVER — CLIQUE $\leq_P$ VERTEX-COVER

: (u,v) $u \in V'$ $v \in V'$ V'. **:** k ?

- **NP:** V' $|V'| = k$, $\rightarrow O(|V|\,|E|)$.

. $G = (V,E)$ $\bar{G} = (V, \bar{E})$, $\bar{E} = \{(u,v) : u \neq v,\ (u,v) \notin E\}$. (G,k) $(\bar{G},\ |V| - k)$ $(O(|V|^2))$.

: G k $\iff$ $\bar{G}$ $|V| - k$.

($\rightarrow$) G k V' . V' "$u, v \in V' \Rightarrow (u,v) \in E$." "$(u,v) \notin E \Rightarrow u \notin V'$ $v \notin V'$", "$(u,v) \in \bar{E} \Rightarrow u \in V - V'$ $v \in V - V'$." $V - V'$ $\bar{G}$ ($|V| - k$).

($\leftarrow$) $\bar{G}$ $|V| - k$ V' . "$(u,v) \in \bar{E} \Rightarrow u \in V'$ $v \in V'$." "$u \notin V'$ $v \notin V' \Rightarrow (u,v) \notin \bar{E}$, $(u,v) \in E$." $V - V'$ G $\rightarrow V - V'$ (k). $\blacksquare$

10 :

NP-

1. $X \in NP$ — + .
2. X NP-hard — NPC Y $Y \leq_P X$.
   - (a) , (b) ($\rightarrow$) , (c) ($\leftarrow$) .
3. ( ) NP-hard .

- : known-hard $\rightarrow$ yours (yours $\leq_P$).
- : yours $\rightarrow$ known-easy (yours $\leq_P$).
- $\leq_P$, .

- $p \rightarrow q \equiv \neg p \vee q$ ( ).
- $P \leftrightarrow Q \equiv \neg P \leftrightarrow \neg Q$ (yes yes no no).
- $(P \rightarrow Q) \equiv (\neg Q \rightarrow \neg P)$ ( ).